SIMATS ENGINEERING

ASSESSMENT TOOL 3 - DEBUGGING & OPTIMISATION TASK

Course Name	Computer Networks
Course Code	CSA0709
Course Outcome (CO2)	Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology (BL3)
Software	Cisco Packet Tracer, Wireshark
Name	Ashmitha K
Register Number	1912260423
Assessment Tool Weightage	30%

Code of Conduct: I, Ashmitha K (1912260423), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Question 1 (10 Marks)

Question: A workstation is configured with IP 192.168.10.65, subnet mask 255.255.255.192 (/26) and default gateway 192.168.10.1, but it cannot communicate with other systems in subnet 192.168.10.0/26. Debug the configuration: identify the subnet of the IP, network/broadcast address and host range, verify the IP and gateway, identify the errors, assign a suitable IP and gateway, and calculate usable hosts after correction.

1. Problem Identification

- The workstation is meant to work in 192.168.10.0/26 but is not able to reach hosts of that subnet.
- Symptom points to a subnet mismatch: the IP address and the gateway may not belong to the same network.

2. Debugging (Step-by-Step)

Step 1 - Subnet of 192.168.10.65/26: block size = 256 - 192 = 64, so subnets are .0, .64, .128, .192. Binary check: 65 = 01000001; AND with 192 (11000000) = 01000000 = 64. The IP belongs to **192.168.10.64/26**.

Step 2 - Details of the two /26 subnets:

Parameter	Configured IP subnet	Intended subnet
Network address	192.168.10.64	192.168.10.0
Broadcast address	192.168.10.127	192.168.10.63
Valid host range	192.168.10.65 - 192.168.10.126	192.168.10.1 - 192.168.10.62
Usable hosts	2^6 - 2 = 62	62

Step 3 - Verify IP and gateway: The IP 192.168.10.65 is a valid host address inside 192.168.10.64/26, but it is outside the intended 192.168.10.0/26. The gateway 192.168.10.1 (1 AND 192 = 0) is in 192.168.10.0/26, i.e. a different subnet from the workstation, so it is not reachable directly.

Step 4 - Configuration errors:

- Error 1: IP address 192.168.10.65 belongs to subnet 192.168.10.64/26, not to the intended 192.168.10.0/26, so hosts of 192.168.10.0/26 appear off-link and traffic goes to the gateway.
- Error 2: Default gateway 192.168.10.1 is in a different subnet from the workstation IP, so the workstation cannot even send packets to its gateway (no ARP reply on the same LAN).

3. Optimization (Corrected Configuration)

Assign an address inside the intended subnet and keep the gateway (first usable address, router interface) in the same subnet:

Setting	Wrong value	Corrected value
IP address	192.168.10.65	192.168.10.10
Subnet mask	255.255.255.192	255.255.255.192 (/26)
Default gateway	192.168.10.1 (different subnet from IP)	192.168.10.1 (now same subnet)

```
! Router interface (Cisco IOS / Packet Tracer)
interface GigabitEthernet0/0
 ip address 192.168.10.1 255.255.255.192
 no shutdown
! Workstation (Desktop > IP Configuration)
IP Address      : 192.168.10.10
Subnet Mask     : 255.255.255.192
Default Gateway : 192.168.10.1
```

4. Analysis and Verification

Usable hosts after correction: 2^6 - 2 = **62** in 192.168.10.0/26 (network .0 and broadcast .63 excluded). One address is used by the router gateway, so 61 remain for workstations.
Verification: In Packet Tracer, ping 192.168.10.1 and another host in 192.168.10.0/26 from the workstation; both replies should succeed. In Wireshark, ARP requests from 192.168.10.10 now get replies on the same LAN, which confirms the subnet fix.

Question 2 (10 Marks)

Question: Device A has IPv6 address 2001:db8::ff00:42:8329/64 and Device B has 2001:db8:0:0:1::1/64. Both are on the same network but communication fails. Expand both addresses, determine the network prefix of each, verify whether they are in the same subnet, identify the addressing/prefix mismatch, optimize the configuration, and calculate the host addresses in the corrected subnet.

1. Problem Identification

- Both devices are physically on the same network, yet communication fails, so the IPv6 addressing (prefix and interface ID) must be checked.
- Both use a /64 prefix, so the first 64 bits (first four groups) decide the subnet.

2. Debugging (Step-by-Step)

Step 1 - Expand Device A: 2001:db8::ff00:42:8329 has 5 visible groups, so "::" = 8 - 5 = 3 zero groups. Full form: **2001:0db8:0000:0000:0000:ff00:0042:8329**.

Step 2 - Expand Device B: 2001:db8:0:0:1::1 has 6 visible groups (2001, db8, 0, 0, 1, 1), so "::" = 2 zero groups. Full form: **2001:0db8:0000:0000:0001:0000:0000:0001**.

Step 3 - Network prefix (/64 = first four groups):

Device	Full address	Network prefix (/64)	Interface ID
A	2001:0db8:0000:0000:0000:ff00:0042:8329	2001:db8::/64	0000:ff00:0042:8329
B	2001:0db8:0000:0000:0001:0000:0000:0001	2001:db8::/64	0001:0000:0000:0001

Step 4 - Same subnet? Yes. Both first-four-group values are 2001:0db8:0000:0000, so both addresses are in **2001:db8::/64**. Therefore the prefix itself does not cause the failure.

Step 5 - Identify the mismatch:

- The apparent mismatch is in Device B's notation: 2001:db8:0:0:1::1 puts a non-zero value (1) in the fifth group, which belongs to the host (interface ID) part, not the prefix. If the engineer meant B to be on a separate subnet 2001:db8:0:1::/64, the address was mistyped (extra '0:' group).
- Interface IDs are irregular and inconsistent (ff00:42:8329 versus 1:0:0:1), which makes errors likely; neither device uses a clear gateway address such as 2001:db8::1.
- Because the layer-3 prefix is identical, the real cause must be checked in the configuration/verification: different prefix length actually applied on one interface, interface or VLAN shut down/mismatched, IPv6 routing not enabled on the router, or ICMPv6/Neighbor Discovery blocked.

3. Optimization (Corrected Configuration)

Assign clean, consistent addresses from the same /64 and keep a gateway on the router:

Device	Corrected IPv6 address	Prefix	Gateway
Router interface	2001:db8::1	/64	-
Device A	2001:db8::a	/64	2001:db8::1
Device B	2001:db8::b	/64	2001:db8::1

```
! Router (Cisco IOS)
ipv6 unicast-routing
interface GigabitEthernet0/0
 ipv6 address 2001:DB8::1/64
 no shutdown
! Device A (static)          ! Device B (static)
IPv6 2001:DB8::A/64          IPv6 2001:DB8::B/64
Gateway 2001:DB8::1          Gateway 2001:DB8::1
```

4. Analysis and Verification

Host addresses in the corrected subnet: host bits = 128 - 64 = 64, so addresses = 2^{64} = **18,446,744,073,709,551,616** (IPv6 has no broadcast address, so none is subtracted).
Verification: Use ping 2001:DB8::B from Device A (ping6 in Packet Tracer) and show ipv6 interface brief on the router. In Wireshark, filter icmpv6: Neighbor Solicitation/Advertisement and Echo Request/Reply between

the two devices confirm that they communicate inside the same /64.

Question 3 (10 Marks)

Question: 192.168.1.0/24 is divided among departments using masks /26, /27 and /28. Some departments report insufficient IP addresses while others have many unused addresses. Identify network, broadcast and host range of each subnet, calculate usable and unused addresses, identify inefficient allocations and mistakes, and recommend an optimized VLSM scheme.

1. Problem Identification

- The block 192.168.1.0/24 (256 addresses) is split into one /26, one /27 and one /28 subnet.
- Problem: the subnet sizes do not match what each department needs - one is oversized (wasted addresses) and one is undersized (shortage).
- The question does not give the departments' host requirements, so the following example requirements are **assumed** for the analysis: Dept A needs 20 hosts, Dept B needs 45 hosts, Dept C needs 12 hosts.

2. Debugging (Step-by-Step)

Step 1 - Existing allocation (given masks assigned in order, A = /26, B = /27, C = /28):

Dept	Subnet	Network	Broadcast	Host range	Usable	Needed (assumed)	Result
A	/26	192.168.1.0	192.168.1.63	.1 - .62	62	20	42 unused (oversized)
B	/27	192.168.1.64	192.168.1.95	.65 - .94	30	45	Short by 15
C	/28	192.168.1.96	192.168.1.111	.97 - .110	14	12	2 spare (OK)

Step 2 - Addresses not in any subnet: 192.168.1.112 - 192.168.1.255 = 144 addresses are unallocated.

Step 3 - Mistakes located:

- Subnet sizes were assigned without sizing them to demand: the largest subnet (/26) went to the department with the smallest need among A and B.
- Dept B's /27 gives only 30 usable hosts against 45 needed (shortage), while Dept A's /26 wastes 42 addresses.
- No separate small subnets are reserved for router-to-router links.

3. Optimization (Corrected Configuration)

VLSM method: for a required host count H choose the smallest prefix with 2^h - 2 >= H; allocate from the largest requirement to the smallest: B (45 hosts) needs /26 (62); A (20 hosts) needs /27 (30); C (12 hosts) needs /28 (14).

Dept	Needed	Prefix	Network	Broadcast	Host range	Usable	Spare
B	45	/26	192.168.1.0	192.168.1.63	.1 - .62	62	17
A	20	/27	192.168.1.64	192.168.1.95	.65 - .94	30	10
C	12	/28	192.168.1.96	192.168.1.111	.97 - .110	14	2
Link 1	2	/30	192.168.1.112	192.168.1.115	.113 - .114	2	0
Link 2	2	/30	192.168.1.116	192.168.1.119	.117 - .118	2	0

Free space after VLSM: 192.168.1.120 - 192.168.1.255 (136 addresses) is kept for future departments.

4. Analysis and Verification

Utilisation: Total usable hosts in the three department subnets = 62 + 30 + 14 = 106. Before optimization only 20 + 30 + 12 = 62 hosts could be served (58%), with Dept B 15 hosts short. After VLSM, all 77 required hosts fit (about 73% of 106), the shortage is removed and spare addresses inside subnets fall from 44 (42 + 2) to 29 (17 + 10 + 2).

Why it works: VLSM fits each subnet to its demand, so addresses are not wasted, no department is starved, and the unused tail of the /24 stays in one continuous block. **Verification:** configure the subnets on the router in Packet Tracer and ping between departments; use Wireshark to confirm traffic uses the correct gateways.

Question 4 (10 Marks)

Question: Device A has IPv6 address 2001:db8::ff00:42:8329/64 and Device B has 2001:db8:0:0:1::1/64. Both are on the same network but communication fails. Expand both addresses, determine the network prefix of each, verify whether they are in the same subnet, identify the addressing/prefix mismatch, optimize the configuration, and calculate the host addresses in the corrected subnet.

1. Problem Identification

- Both devices are physically on the same network, yet communication fails, so the IPv6 addressing (prefix and interface ID) must be checked.
- Both use a /64 prefix, so the first 64 bits (first four groups) decide the subnet.

2. Debugging (Step-by-Step)

Note: Question 4 repeats the device configuration and tasks of Question 2; the same debugging method and results apply.

Step 1 - Expand Device A: 2001:db8::ff00:42:8329 has 5 visible groups, so "::" = 8 - 5 = 3 zero groups. Full form: **2001:0db8:0000:0000:0000:ff00:0042:8329**.

Step 2 - Expand Device B: 2001:db8:0:0:1::1 has 6 visible groups (2001, db8, 0, 0, 1, 1), so "::" = 2 zero groups. Full form: **2001:0db8:0000:0000:0001:0000:0000:0001**.

Step 3 - Network prefix (/64 = first four groups):

Device	Full address	Network prefix (/64)	Interface ID
A	2001:0db8:0000:0000:0000:ff00:0042:8329	2001:db8::/64	0000:ff00:0042:8329
B	2001:0db8:0000:0000:0001:0000:0000:0001	2001:db8::/64	0001:0000:0000:0001

Step 4 - Same subnet? Yes. Both first-four-group values are 2001:0db8:0000:0000, so both addresses are in **2001:db8::/64**. Therefore the prefix itself does not cause the failure.

Step 5 - Identify the mismatch:

- The apparent mismatch is in Device B's notation: 2001:db8:0:0:1::1 puts a non-zero value (1) in the fifth group, which belongs to the host (interface ID) part, not the prefix. If the engineer meant B to be on a separate subnet 2001:db8:0:1::/64, the address was mistyped (extra '0:' group).
- Interface IDs are irregular and inconsistent (ff00:42:8329 versus 1:0:0:1), which makes errors likely; neither device uses a clear gateway address such as 2001:db8::1.
- Because the layer-3 prefix is identical, the real cause must be checked in the configuration/verification: different prefix length actually applied on one interface, interface or VLAN shut down/mismatched, IPv6 routing not enabled on the router, or ICMPv6/Neighbor Discovery blocked.

3. Optimization (Corrected Configuration)

Assign clean, consistent addresses from the same /64 and keep a gateway on the router:

Device	Corrected IPv6 address	Prefix	Gateway
Router interface	2001:db8::1	/64	-
Device A	2001:db8::a	/64	2001:db8::1
Device B	2001:db8::b	/64	2001:db8::1

```
! Router (Cisco IOS)
ipv6 unicast-routing
interface GigabitEthernet0/0
 ipv6 address 2001:DB8::1/64
 no shutdown
! Device A (static)          ! Device B (static)
IPv6 2001:DB8::A/64          IPv6 2001:DB8::B/64
Gateway 2001:DB8::1          Gateway 2001:DB8::1
```

4. Analysis and Verification

Host addresses in the corrected subnet: host bits = 128 - 64 = 64, so addresses = 2^{64} = **18,446,744,073,709,551,616** (IPv6 has no broadcast address, so none is subtracted).

Verification: Use ping 2001:DB8::B from Device A (ping6 in Packet Tracer) and show ipv6 interface brief on the router. In Wireshark, filter icmpv6: Neighbor Solicitation/Advertisement and Echo Request/Reply between

the two devices confirm that they communicate inside the same /64.

Question 5 (10 Marks)

Question: A router has the routing table: 192.168.1.0/24 via 10.0.0.2; 192.168.2.0/24 via 10.0.0.3; default route via 10.0.0.1. Packets for 192.168.2.100 do not reach the intended network. Debug the routing: find the longest-prefix match and next hop, verify the destination subnet, identify errors, optimize the table, and give the route used if there is no match.

1. Problem Identification

- Destination 192.168.2.100 should reach the 192.168.2.0/24 network, but packets are not arriving.
- The routing table has two specific /24 routes and one default route; the router forwards using the longest-prefix match.

2. Debugging (Step-by-Step)

Step 1 - Does the destination belong to 192.168.2.0/24? Yes. 192.168.2.100 AND 255.255.255.0 = 192.168.2.0; the range is 192.168.2.0 - 192.168.2.255 (hosts .1 - .254), so .100 is a valid host.

Step 2 - Longest-prefix match:

Table entry	Matches 192.168.2.100?	Prefix length
192.168.1.0/24 via 10.0.0.2	No (third octet 1)	-
192.168.2.0/24 via 10.0.0.3	**Yes**	/24
Default route via 10.0.0.1	Yes (matches everything)	/0

The /24 entry is the longest match, so the router selects **next hop 10.0.0.3**; the default route (shortest prefix) is not used.

Step 3 - Errors / possible causes: the routing entry itself is correct for the destination, so the fault is in how the route is set up or in the next-hop path:

- Next hop 10.0.0.3 may be wrong, down or unreachable (the next hops of the 192.168.1.0/24 and 192.168.2.0/24 entries may have been swapped, so 192.168.2.x is sent to the wrong router).
- The next-hop router may have no connected network or return route for 192.168.2.0/24, or the exit interface is not specified/up.
- The default route is written as 'Default Route' without an explicit 0.0.0.0/0 entry, and the subnet mask of the 10.0.0.x link is not stated, so reachability of the next hops cannot be confirmed.

3. Optimization (Corrected Configuration)

The two /24 networks have different next hops and cannot be summarized into one /23 (192.168.1.0/24 and 192.168.2.0/24 are not an aligned pair). The table is optimized by fixing the entries and making each route explicit:

```
! Verify next-hop reachability first
ping 10.0.0.3
show ip route
! Corrected static routes (Cisco IOS)
ip route 192.168.1.0 255.255.255.0 10.0.0.2
ip route 192.168.2.0 255.255.255.0 10.0.0.3
ip route 0.0.0.0 0.0.0.0 10.0.0.1
```

4. Analysis and Verification

Route used if no match exists: the default route **0.0.0.0/0 via 10.0.0.1** (gateway of last resort).
Verification: Run show ip route to confirm the entries and traceroute 192.168.2.100 to see that the first hop is 10.0.0.3. In Wireshark, check that ICMP Echo Requests leave toward 10.0.0.3 and that Echo Replies come back; if no reply arrives, the issue lies at the next-hop router or the destination network, not in this routing table.